

Project Java Report

May 1, 2025

1 Overview

This project implements arbitrary-precision arithmetic operations for integers and floating-point numbers using custom data structures. It avoids reliance on native types for computations and instead defines operations character-by-character using string manipulation.

2 Design Decisions

2.1 Package Structure

The project defines a Java package called `arbitraryarithmetic` that contains two main classes: `Ainteger` and `AFloat`, used to handle large integer and floating-point arithmetic, respectively.

2.2 Class Responsibility

- **Ainteger:** Handles large integer arithmetic. Operations implemented include `add`, `subtract`, `multiply`, and `divide`, all operating on string representations.
- **AFloat:** Handles large floating-point arithmetic using the `Ainteger` class to handle the numerical core of multiplication and division.
- **MyInfArith:** The entry point to the application that parses command-line arguments and delegates operations to either `Ainteger` or `AFloat`.

2.3 Build and Execution

The project uses Apache Ant for build automation, with tasks defined in `build.xml` to compile, jar, and run the application. A `Dockerfile` and a Python helper script further streamline compilation and execution.

3 UML Class Diagrams

3.1 Ainteger Class Diagram

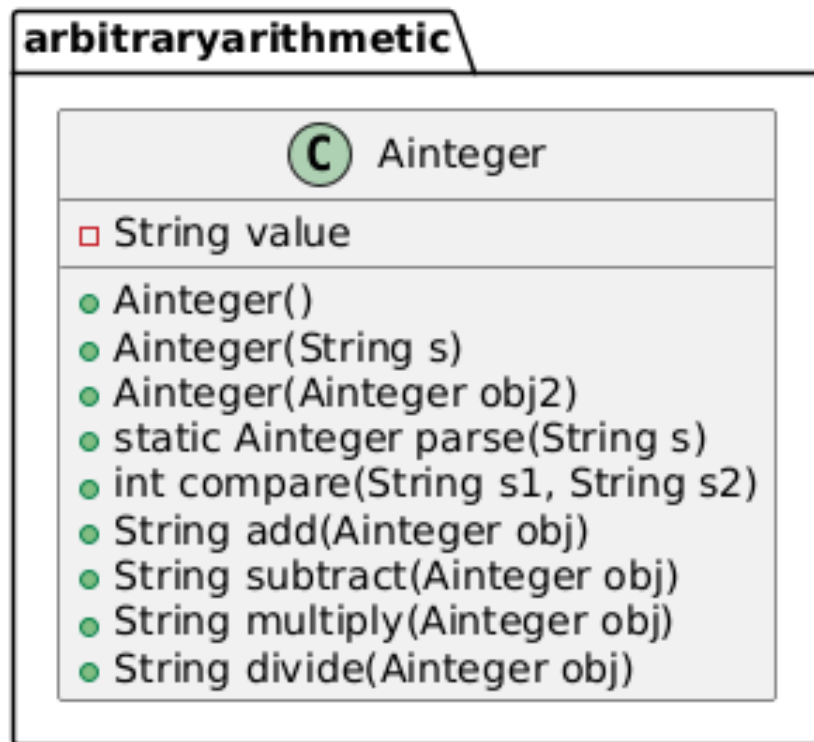


Figure 1: UML for Ainteger

3.2 AFloat Class Diagram

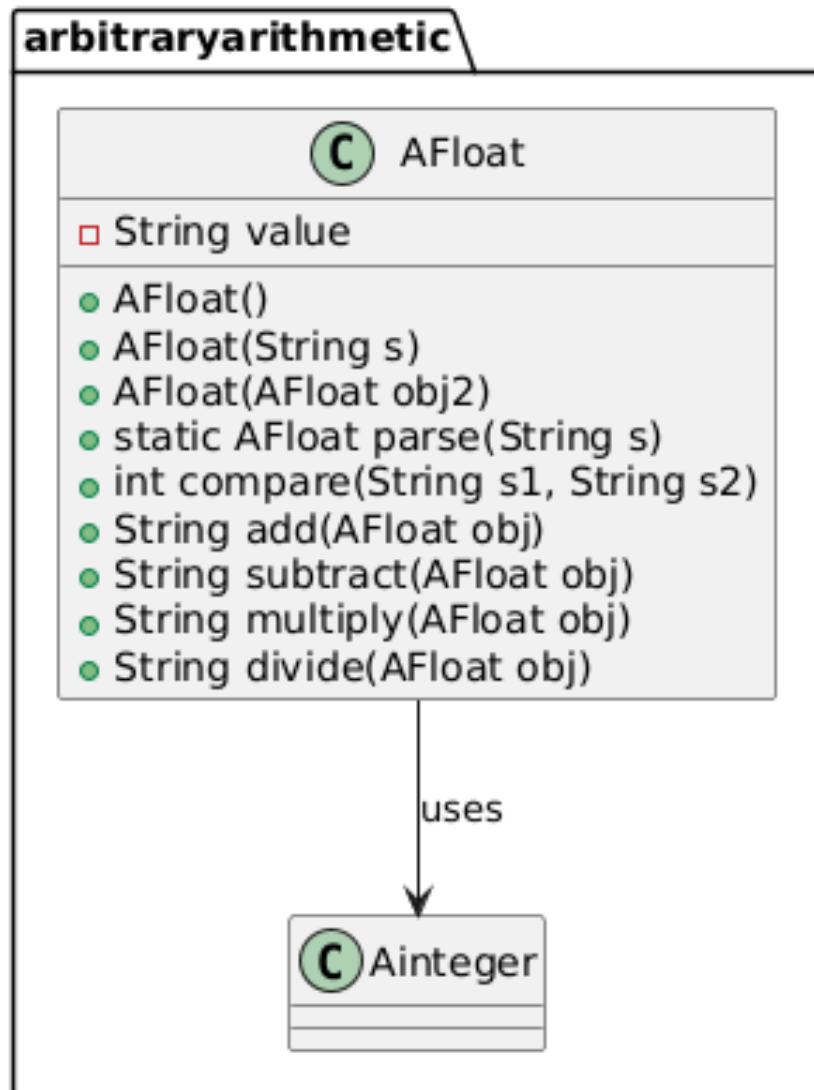


Figure 2: UML for AFloat

4 Implementation Highlights

- All arithmetic operations avoid use of Java's built-in numeric types and instead operate at the character level.
- **AFloat** handles decimal alignment and normalization before performing arithmetic.
- **StringBuilder** is used for efficient character appends during operations.
- The **compare** function in both classes determines numeric precedence using digit-wise comparison.

5 Conclusion

The design emphasizes modular arithmetic logic, clarity, and correctness in computations that exceed standard type boundaries. The architecture allows extension to even more types such as arbitrary-precision complex numbers.